Planning:

Workflow:

- Activities that transform input/output
- Describes work that must be done, who or what does it, what order, under what conditions does it succeed, fail, is rejected or requires escalation.

Workflow management system:

- Stores workflow definitions and creates and manages running workflow instances from those definitions.
- May need to coordinate work items, participants, routine decisions, approvals, events, dependencies, external systems, history, monitoring and reporting.

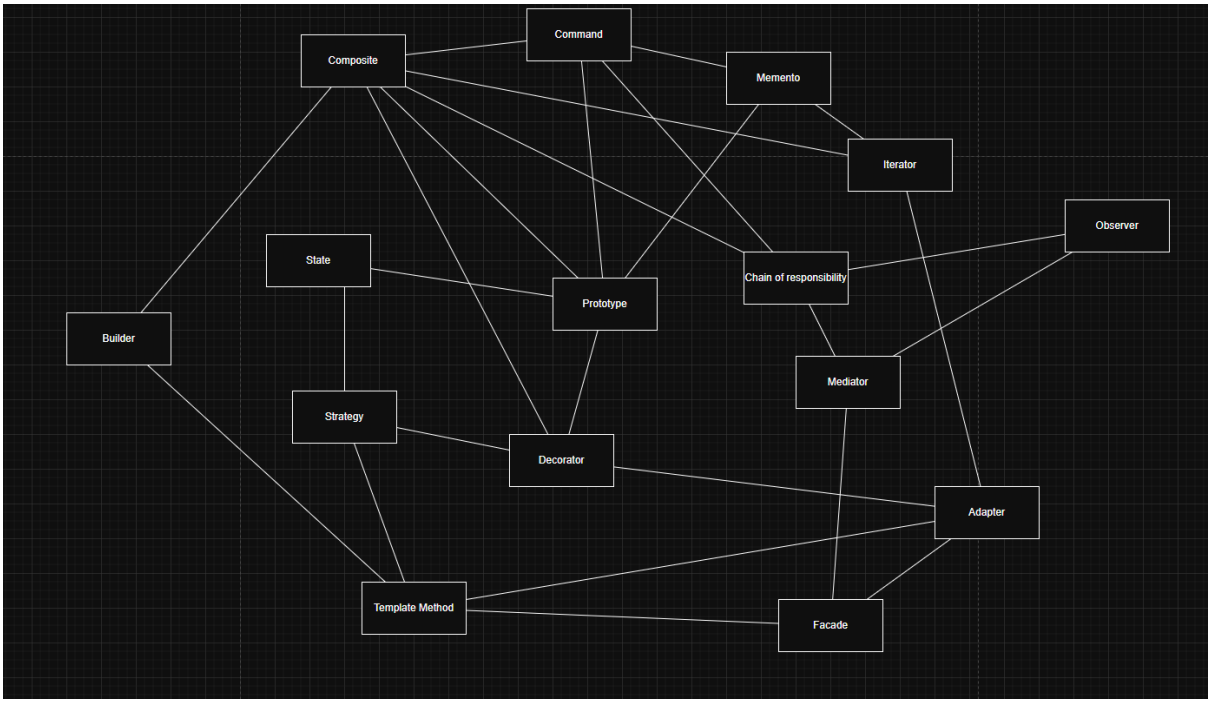
Possible concepts that may appear in a workflow system include:

- workflow definitions and running workflow instances;
- work items, stages and sub-processes;
- participants, roles and responsibilities;
- work-item lifecycles;
- routing and assignment rules;
- approvals, rejection, escalation and deadlines;
- events and triggers;
- data payloads, artefacts, constraints and dependencies;
- workflow construction and execution;
- actions, history and rollback;
- integrations and connectors;
- workflow versioning and traversal;
- access control and resource coordination; and
- monitoring, analytics and reporting.

Prac 2: (research and initial design phase)

Pattern	Use
State	A work item may be created, made available, assigned, started, completed, rejected, cancelled or escalated
Decorator	a task may acquire additional behaviour such as validation, priority handling, auditing, notifications or security without becoming a completely different kind of task.
Prototype	A change to one running instance should therefore not accidentally change every other instance created from the same definition.
Composite	how individual pieces of work and groups of work are represented. (work contains stages, items and sub-processes)
Iterator	navigated without requiring the rest of the system to know their internal storage structure
Strategy	One organisation may assign the next task based on role, another on workload, and another on a rule calculated from workflow data.
Chain of responsibility	Approval may involve one person, several levels, or escalation when nobody responds.
Observer	workflow system must also react to what happens around it. Completing one activity may make another available, an external event may trigger new work, and several interested parts of the system may need to react to the same event.
Adapter	need to communicate with external systems whose interfaces were not designed for it.
Command	support undo, retain actions and state changes
Memento	rollback where sensible, preserve the meaning of older workflow definitions
Template Method	Workflow execution template
Facade	Gives the program a simple starting interface due to the size and complexity of the whole system
Mediator	Handles the communication between classes that reference/wait for other classes specifically
Builder	Assembles workflow definitions from stages, tasks, rules and dependencies

Pattern relations(From pdf notes):



Task 1 Prep:

Web links and questions answered:

[RMV1-16.PDF (WfMC Workflow Reference Model)]:

- What is a workflow and a workflow management system?
- How do workflow systems interact with external systems? (the interfaces between the engine and other tools)
- How are participants, roles and responsibilities represented? (partial)
- How are running workflows monitored and reported on? (partial, via the monitoring interface)
- What makes a workflow engine reusable across processes? (indirect, through the generic components and interfaces)

[Workflow Management Coalition Glossary and Terminology]:

- What is a workflow and a workflow management system?
- What is the difference between a workflow definition and a running instance?
- Which information belongs to a definition and which to an instance?
- How are activities or work items represented?
- How do real systems model nested activities or sub-processes?

Workflow Patterns:

- How are dependencies handled? (sequence, parallelism, synchronisation)
- How is work routed? (choice and splits, but not assignment to people)
- What makes an engine reusable? (partial, because it shows common constructs across products)

Events and Event History | Temporal Documentation:

- How is workflow history or audit information recorded?
- How are events and failures handled?

What is a Temporal Retry Policy? | Temporal Documentation:

- How are failures handled?
- Which decisions are normally configurable rather than hard-coded in the engine?

Application failures | Temporal Documentation:

- How are failures handled? (failure types such as timeout, cancelled and child workflow)
- How do nested activities fail? (partial, through child workflow failures)

Process Versioning | docs.camunda.org:

- How are changes to definitions handled while work is in progress?
- Which information belongs to a definition and which to an instance? (instances stay linked to the version they started on)

Versioning process definitions | Camunda 8 Docs:

- How are changes to definitions handled while work is in progress? (migrate or don't migrate)
- How is history preserved across versions? (migration keeps the instance's identity and audit trail)

Leiden University thesis chapter on BPMN 2:

- How are escalation and events handled? (escalation events, plus error, timer and cancel events)
- How do real systems model nested activities or sub-processes? (sub-processes, event sub-processes and call activities)
- What is a process instance? (a process can have many instances)

Still need:

- How are approvals, rejection and escalation handled?